MILITARY SPECIFICATION
REMOVER, AIRCRAFT CANOPY, M8A1
REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, M8A1 Remanufacture Metal Parts Assembly and modification requirements to incorporate the new load limiting device in the gear box assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal
- QQ-P-416 - Plating, Cadmium (Electro Deposited)

Military
- MIL-A-2550 - Ammunitions and Special Weapons, General Specification for
- MIL-C-5541 - Chemical Films and Chemical Film Materials for Aluminum Alloys
- MIL-L-46010 - Lubricant, Solid Film: Heat Cured, Corrosion Inhibiting

STANDARDS

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command
- F10522121 - Remover, Aircraft Canopy, M8A1, Metal Parts Assembly

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DRAWINGS (Cont'd)

Munitions Command

C8837890	- Carton, Packing, Ammunition for Remover, Aircraft Canopy M8 or M8A1 with Cartridge M150
C9240994	- Carton, Alternative Packing, Ammunition for (1) Remover, Aircraft Canopy M8 or M8A1 with Cartridge M150
C9230161	- Box, Packing, Ammunition for (1) Remover, Aircraft Canopy M8 or M8A1, with Cartridge M150
IEL-10522121	- Inspection Equipment List for Remover, Aircraft Canopy M8A1 Metal Parts Assembly

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI) 1430 Broadway, New York, N. Y. 10018).

3. REQUIREMENTS

3.1 General - Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Hydrostatic pressure - The outside tube, C10522032 and inside tube C10522036 shall withstand a hydrostatic pressure of 2500 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.2).

3.3 Manual function - Without electrical power and with the clutch disengaged, the pull force required to extend the inside tube assembly shall not exceed 40 pounds.

3.4 Reversibility load - The remover shall be capable of withstanding a load of 2500 pounds minimum in tension and 1740 pounds minimum in compression at a mid-stroke of approximately 5 inches. The remover shall not travel or unlock due to the applied load.

3.5 Operating performance - The remover shall be cycled and shall be capable of operating to the loads described by the load curves A, B and C of Figure 1 (see 6.3) and shall comply with the following:

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3.5.1 The motor running current drain under load curve B of Figure 1 shall not exceed 13 amperes at 26 to 30 volts D.C. at ambient conditions.

3.5.2 The motor running current drain under load curve C of Figure 1 shall not exceed 20 amperes at 26 to 30 volts D.C. at ambient conditions.

3.5.3 Time required for linear extension of the remover for a normal stroke of 9.38 inches and under load curve B of Figure 1 shall not exceed 12 seconds using the power requirement specified in 3.5.1.

3.5.4 Time required for linear extension of the remover for a normal stroke of 9.38 inches and under curve C of Figure 1 shall not exceed 15 seconds using the power requirement specified in 3.5.2.

3.5.5 Time requirement for linear retraction of the remover for a normal stroke of 9.38 inches under curve A of Figure 1 shall not exceed 15 seconds at 26 to 30 volts D.C. with a running current drain not to exceed 15 amperes at ambient conditions

3.5.6 When the load aids the operation of the remover for the full stroke of 9.38 inches, the operating time shall be 5 seconds minimum to 10 seconds maximum and the running current shall not exceed 10 amperes at 26 to 30 volts D.C.

3.5.7 There shall be no structural failure of any component under the inertia loads imposed by the sudden interruption of electrical power.

3.5.8 At the end of the duty cycle for normal load curve B, Figure 1, the remover shall pull against the stop with a force, known as cinch-down load, of 1000 to 1450 pounds. Under this cinch-down load requirement, the load limit switch shall de-energize the remover electric motor.

3.5.9. Time required to engage clutch shall not exceed 500 milliseconds (see 6.4).

3.5.10 After the remover has cinched down, the clutch shall be disengaged. The time required to disengage the clutch shall not exceed 500 milliseconds (see 6.5).

3.5.11 The remover shall not fail to operate under normal load curve B of Figure 1 at a voltage imput of 18 volts D.C.

3.6 The Gear Box and Motor Mount Assembly which were manufactured to the original drawing FF9838 shall be retained as a matched set.

3.6.1 The remover metal parts assembly shall contain the following new parts. Those parts which are to be replaced are considered 100% throw-away items.

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NAME OF PART	DWG. NO.
Sleeve (Bulk Material)	MIL-I-631
Actuator Switch (3 req'd)	C10522135
Detent Assembly	B10522055
"O" Ring (2 req'd)	MS28775-008
Pin Shear	A10522065
Plug, Shipping	A8593299
Screw, Shear (2 req'd)	B10522062
Screw	B10522061
Spring, Detent Level	FA 29923
Spring	B10522090
Spring, Piston (2 req'd)	B10522068
Spring, Clutch	B10522110
Packing "O" Ring	MS 9021-031
Packing "O" Ring	AN6227-7
Packing "O" Ring	MS28775-013
Packing "O" Ring	MS28775-332
Packing "O" Ring	MS29513-127
Packing "O" Ring	MS29513-134
Packing "O" Ring	MS29513-142
Packing "O" Ring	MS29513-145
Clamp, Cushion Support (2 req'd)	MS21919-H4
Spring, Helical Extension	B10522095
Bushing (for adapter)	AN3420-8
Cap, Plastic Dust	MS25177-22
Nut	21083-C6
Pin, Spring	MS16562-50
Pin, Spring	C10543076-3
Pin, Spring	MS16562-47
Pin, Spring	MS9048-076
Pin, Spring	MS16562-30
Pin, Double Roll	MS9048-164
	10543076-2
Pin, Spring	MS16562-43
Pin, Spring	MS16562-29
Screw, Set	MS51693-20
*Key (2)	B10522037
**Detent, Worm (2 req'd)	C10522152
**Detent, Shaft	C10522153
**Spring (3 req'd)	MS24585-50

*Replace only to update remover to latest configuration.

**Replace only after the modification has been accomplished.

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3.6.2 Any spring pin removed from an assembly or if found defective shall be replaced with a new spring pin.

3.6.3 Bolts, nuts, screw inserts or other threaded components which show evidence of cross threading, stripping or excessive wear shall be replaced.

3.6.4 Bearing, ball bearing screw assemblies, and ball bearings which show evidence of roughness, binding or damage shall be replaced.

3.6.5 Switches which contain a loose plunger, damaged wiring, gummy or defective plunger seals shall be replaced.

3.6.6 Electric motor shall be replaced if it is unreparable.

3.6.7 Any gear with excessive back lash, worn or broken teeth or improper mating, shall be replaced with a new gear.

3.6.8 Gear Box and Ballistic Assembly - Any part in the Gear Gox and Ballistic Assembly which shows evidence of distortion, excessive wear, looseness, or any other damage shall be repaired or replaced with a new part.

3.6.9 Gear Box lubrication - The grease in the Gear Box cavity shall be removed and replaced with fresh grease conforming to the requirements of the applicable grease specification.

3.7 Protective finish - Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configurations. All remanufactured exterior components and components undergoing modification shall have the existing protective finish stripped and the specified protective finish re-applied. (See 6.6).

3.8 Workmanship - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects - Components shall be free from cracks, splits or other metal defects.

3.8.2 Burr - No part shall have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.8.3 Foreign matter - Chips, dirt, grease (except where required), rust, corrosion, or other foreign matter shall be removed.

3.9 Metal Parts Assembly - Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the following:

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3.9.1 The modified metal parts assembly shall contain the following newly designed components and sub-assemblies (See 6.7.2).

NOMENCLATURE	OLD DWG NO.	NEW PROD NO.	NO. REQ'D
Gear Clutch Ass'y		C10522155	-
Shims		B10543075	1
Detent, Worm		C10522152	2
Detent, Shaft		C10522153	1
Retainer		C10522154	1
Worm Shaft Ass'y	C10522091	C10522157	-
Spring	B10522090	C10522156	1
Spacer		B10543073	1
Spring		MS24585-50	3
Pin, Spring		MS16562-24	2
Insert, Screw Locking		MS21209-C420	2
Screw		MS51957-17	2

3.9.2 The modified metal parts assembly shall contain the following components which shall be modified to the applicable modification drawing:

NOMENCLATURE	ORIG NO.	MOD DWG.	NEW PRODUCTION DWG. (See 3.9.2.1)	REQ'D
Gear Box Ass'y	F10522096	-	Rev. B.	-
Worm	C10522083	C10522148	C10522139	1
Tube, Inner*	D10522036	D10522036	Rev. A.	1
Gear, Bevel	C10522085	B10522150	C10522151	1
Shaft, Worm	C10522084	C10522146	C10522147	1

*Note: Strip Emralon coating and degrease chrome plating. Apply solid film lubricant in accordance with appropriate drawing.

3.9.2.1 New production parts shall be in accordance with the applicable new production drawing.

3.9.2.2 The gear box and motor mount assembly which was manufactured to the original drawing FF9838 shall be retained as a matched set.

3.9.2.3 In the modification of the M8A1 Remover to the M8A1 with the new load limiting device in the gear box assembly, the following components shall be deleted and discarded: (See 6.7.2).

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NOMENCLATURE	DWG. NO.	NO.
Spring	B10522090 (See 3.9.1)	1
Hub	B10522086	1
Rivet	MS20470	3
Pin, Spring	MS16562-35	3
Pin, Spring	MS16562-33	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define the Quality Assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions

4.2.1 Lot

4.2.1.1 Submission of product - The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot - For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 removers, maximum, exclusive of test samples, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination - The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts - Examination of resuable parts, which require no reprocessing shall be limited to inspection of defects that require visual type defects. All nonconforming parts shall be rejected.

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4.2.2.2 Reprocessed parts - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables I and II. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected.

TABLE I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE II

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over - 48	8	11	17	20

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4.2.2.3 Head, Firing Drawing B10522045 covering a detail of Drawing F10522121

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter of small inside thread	Gage
102. Diameter of large inside thread	Gage
103. Diameter of long bore	Gage
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Tube, Outside Assembly, Drawing C10522032 covering a detail of Drawing B10522121

CRITICAL	
None defined	
MAJOR	
101. Diameter of external thread	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Ring, Locking, Drawing B10522063 covering a detail of Drawing F10522121

CRITICAL	
None defined	

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4.2.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of inside thread	Gage
102. Metal defective	Visual
103. Protective finish inadequate	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.6 Cover, Drawing B10522140 covering a detail of Drawing F10522121

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 End, Rod, Drawing C10522049 covering a detail of Drawing F10522121

CRITICAL	
None defined	
MAJOR	
101. Diameter of outside thread	Gage
102. Diameter of bore	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual

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4.2.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	
4.2.2.8 Plate, Cover, Drawing B10522024 covering a detail of Drawing F10522121	
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	
4.2.2.9 Retainer, Bearing, Drawing B10522050 covering a detail of Drawing F10522121	
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.10 Housing, Left Bearing, Drawing B10522088 covering a detail of Drawing B10522121

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of small diameter, min.	Gage
102.	Diameter of counterbore, max.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.11 Plug, Inside Tube, Drawing C10522041 covering a detail of Drawing F10522121

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of inside thread	Gage
102.	Diameter of outside thread	Gage
103.	Diameter of spring pin hole	Gage
104.	Spring pin holes, unplugged	Visual
105.	Metal defective	Visual
106.	Protective finish inadequate or defective	Visual
107.	Burr	Visual
108.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.12 Cover, Drawing B10522016 covering a detail of Drawing F10522121

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.13 Ring, Shear, Drawing C1052048 covering a detail of Drawing F10522121

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Outside diameter	Gage
102. Metal Defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.14 Housing, Main Bearing, Drawing C10522039 covering a detail of Drawing F10522121

Defect	Method of Inspection
CRITICAL	
None defined	

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4.2.2.14 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of large bore, max.	Gage
102. Diameter of small bore	Gage
103. Diameter or large outside shoulder, min.	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
None defined	

4.2.2.15 Adapter Housing, Drawing D10543246 covering a detail of Drawing F10522121

CRITICAL	
None defined	
MAJOR	
101. Diameter of inside thread	Gage
102. Diameter of outside thread	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Metal defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
108. Pellet missing, loose or improperly installed	Visual
MINOR	
None defined	

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4.2.2.16 Ball Bearing, Screw Assembly, Drawing 10522056 covering a detail of Drawing F10522121

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Any part missing	Visual
102.	Modification missing or incorrect	Visual
103.	Protective finish inadequate or defective	Visual
104.	Metal defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.17 Mount, Motor, Drawing D10522001 covering a detail of Drawing F10522121

CRITICAL		
	None defined	
MAJOR		
101.	Diameter of plug hole incorrect	Gage
102.	Diameter of plug incorrect	Gage
103.	Plug not flush with surface	Gage
104.	Metal defective	Visual
105.	Protective finish inadequate or defective	Visual
106.	Burr	Visual
107.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.18 Drum, Clutch, Drawing C10522042 covering a detail of Drawing F10522121

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter of counterbore incorrect	Gage
102. Metal defective	Visual
103. Protective finish inadequate	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.19 Cup, Drawing C10522043 covering a detail of Drawing F10522121

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.20 Race, Outer Drawing B10522077 covering a detail of Drawing F10522121

CRITICAL	
None defined	

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4.2.2.20 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.21 Lever, Drawing B10543079 covering a detail of Drawing F10522121

CRITICAL	
None defined	
MAJOR	
101. Surface finish incorrect	Visual
102. Metal defective	Visual
103. Protective finish inadequate	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.22 Box, Gear, Drawing D10522006 covering a detail of Drawing F10522121

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.23 Shaft, Worm (Modification), C10522146 covering a detail of Drawing F10522121

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of diameter on left side of flange to large diameter on right side of flange	Gage
102.	Concentricity of small diameter to large diameter on right side of flange	Gage
103.	True position of flange diameter to large diameter on right side of flange	Gage
104.	Diameter of flange	Gage
105.	Width of flange	Gage
106.	Length from face on right view of flange to face of large diameter	Gage
107.	Length from face on left view of flange to back face of small diameter.	Gage
108.	Diameter of small diameter	Gage
109.	Diameter of diameter on left side of flange after plating	Gage
110.	Angular location of face of groove to centerline of keyway	Gage
111.	Location of face of groove to outside diameter	Gage
112.	Width of groove	Gage
113.	Radius at bottom of groove	Gage
114.	Angle of groove	Gage
115.	Profile of groove radius	Gage
116.	Depth of face of groove	Gage
117.	Location of groove from face of flange	Gage
118.	Surface finish improper	Gage
119.	Metal defective	Visual
120.	Protective finish inadequate or defective	Visual
121.	Radii or chamfers missing or incorrect	Visual
122.	Burr or sharp edge	Visual
123.	Foreign matter	Visual
MINOR		
	None defined	

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4.2.2.24 Worm (Modification) C10522148 covering a detail of Drawing F10522121

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of counterbore to inside diameter	Gage
102.	Depth of counterbore	Gage
103.	Location of slot to centerline of keyway	Gage
104.	Width of slot	Gage
105.	Depth of slot	Gage
106.	Surface finish improper	Gage
107.	Metal defective	Visual
108.	Productive finish inadequate or defective	Visual
109.	Burr or sharp edge	Visual
110.	Radii missing	Visual
111.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.25 Gear, Bevel (Modification) B10522150 covering a detail of Drawing F10522121

CRITICAL		
	None defined	
MAJOR		
101.	True position of bore to hub bore	Gage
102.	True position or large holes to bore	Gage
103.	True position of intermediate size holes to bore	Gage
104.	Diameter of bore	Gage
105.	Diameter of large holes	Gage
106.	Diameter of intermediate size holes	Gage
107.	Location of large holes	Gage
108.	Location of small holes	Gage

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4.2.2.25 (Cont'd)

Defect	Method of Inspection
MAJOR	
109. Diameter of spotface, max.	Gage
110. Depth of spotface, max.	Gage
111. Metal defective	Visual
112. Protective finish inadequate or defective	Visual
113. Surface finish improper	Gage
114. Burr or sharp edge	Visual
115. Foreign matter	Visual
MINOR	
None defined	

4.2.2.26 Tube, Inner D10522036, Rev. A, covering a detail of Drawing F10522121

CRITICAL	
None defined	
MAJOR	
101. Diameter of inside thread	Visual
102. Metal defective	Visual
103. Solid film coating inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.27 Metal Parts Assembly, Drawing F10522121

CRITICAL	
1. Length of firing pin striker	Gage
2. Improper or insecure assembly	Manual and Visual
3. Loose or improper wiring	Manual and Visual
4. Improper soldering	Visual
5. Gas port hole obstructed	Visual
6. Any part missing	Visual
7. Marking missing, incorrect or illegible	Visual
8. Burr	Visual
9. Foreign matter	Visual

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4.2.2.27 (Cont'd)

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
None defined	

4.2.2.28 Packing in Inner Box, Drawing C8837890

CRITICAL		
None defined		
MAJOR		
None Defined		
MINOR		
201.	Contents loose	Manual
202.	Top of box loose	Manual
203.	Strapping missing, loose or improperly assembled	Manual and Visual
204.	Supports or fillers missing from box	Visual
205.	Box damaged	Visual
206.	Seal missing or improperly positioned	Visual
207.	Motor wires loose	Visual
208.	Marking missing, incorrect or illegible	Visual

4.2.2.29 Packing in Outer Box, Drawing C9230161

CRITICAL	
None defined	
MAJOR	
None defined	

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4.2.2.29 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Liner, fillers, cushings or spacers missing	Visual
203. Strapping missing, loose or improperly assembled	Visual
204. Hardware missing, loose or improperly engaged	Visual
205. Box damaged	Visual
206. Contents of box exposed	Visual
207. Car seal missing or improperly sealed	Visual
208. Marking missing, incorrect or illegible	Visual

4.2.2.30 Carton (Outer) Alternative Packing, Drawing C9240994

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Box broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete	Visual

4.2.3 Inspection - The following tests shall be performed in accordance with 4.3.

4.2.3.1 Hydrostatic pressure - Each outside tube shall be subjected to the hydrostatic pressure test. Failure of any item to comply with the requirements of 3.2 shall be cause for rejection of the item.

4.2.3.2 Protective finish, replated components - The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standards listed on the drawing.

4.2.3.3 Manual function - Each remover shall be tested for the pull force required to extent the inside tube. Failure of any remover to comply with the requirements of 3.3 shall be cause for rejection of the remover.

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4.2.3.4 Reversibility load - Each remover shall be tested for reversibility load. Failure of any remover to comply with the requirements of 3.4 shall be cause for rejection of the remover.

4.2.3.5 Operating performance - Each remover shall be cycled electrically 5 times each to the load curves of Figure 1. Failure of the remover to comply with the applicable requirements of 3.5 shall be cause for rejection.

4.2.4 Inspection equipment - The examination and tests shall be made using equipment listed on IEL - 10522121.

4.3 Test methods and procedures

4.3.1 Hydrostatic pressure - Prior to the application of protective finish the outside tube and inside tube shall be mounted in the test device and subjected to an internal hydrostatic pressure to determine compliance with the requirements of 3.2. The test device used shall internally seal the tube at the prescribed seal surfaces without applying an axial load other than that applied by the liquid pressure.

4.3.2 Protective finish, replated components - Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.3.3 Operating performance - The remover shall be secured in the test fixture between remover mounting points and tested in accordance with 4.2.3.5. The indicated values obtained shall be recorded and used to determine compliance with 3.5.

4.3.4 Manual function - The inside tube shall be manually extended by a pull force, with the clutch disengaged and without electric power as specified in 3.3. The pull force shall be recorded.

4.3.5 Reversibility load - The test shall be performed at zero inches, midpoint of stroke and at full stroke with the load in tension and compression as specified in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Packing and marking shall be as specified by the procureing activity. Level A - Drawings C8837890 and C9230161, Level B Drawings C9240994 and C9230161.

6. NOTES

6.1 Ordering data - Procurement documents should spcify the title, number and date of this specification.

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6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 A complete electrical operating cycle consists of clutch engagement, extension, retraction, cinch-down and clutch disengagement.

6.4 Clutch engagement occure with the closing of the up-switch on the control box. The failure to engage the clutch is detected with no cinchdown.

6.5 Clutch disengage time is taken from energizing the disengage solenoid and releasing the cinch-down load to current shut off at disengage solenoid.

6.6 Reprocessed exterior components are listed in Table III.

TABLE III

Name of Part	Drawing Number
Ballistic Assembly	
Adapter Housing	D10543246
Head, Firing	D10522045
Plug, Inside Tube	C10522041
Ring, Locking	B10522063
Ring, Shear	C10522048
Rod, End	C10522049
Gear Box Assembly	
Cover	B10522140
Cover	B10522016
Cover, Plate	B10522024
Housing, Left Bearing	B10522088
Retainer, Bearing	B10522050

6.6.1 The outside surfaces of the remanufactured engage and disengage solenoids shall be cleaned with steel wool and a coating of acrylon shall be sprayed on the outside diameter.

6.6.2 Bare spots on the electrical motor shall be touch-up with black paint.

6.6.3 Exposure of base metal on all other internal aluminum components shall be treated with a chemical film, Grade A or B, Class 1, Specification MIL-C-5541.

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6.6.4 Nylon pellets - Components which contain nylon pellets and are to be replated shall have the pellets removed and new pellets installed after plating in accordance with the applicable drawing.

6.6.5 Locking inserts - Components which contain locking inserts and are to be replated shall have the locking inserts removed and new inserts installed after plating in accordance with the applicable drawing.

6.7 Special instructions covering repair and assembly of remanufactured components are as follows:

6.7.1 Plug - (C10522041) - Plugs which contain elongated or damaged spring pin hole on plug boss may be reclaimed by locating an additional hole approximately 45 degrees from the existing hole. Number of holes in plug shall be limited to three (3). The old holes shall be plugged with plastic steel compound (DEVCON).

6.7.2 When the load limiting modification has been incorporated into the gear box assembly, paragraphs 3.9.1, 3.9.2 and 3.9.2.3 do not apply.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1377-A513

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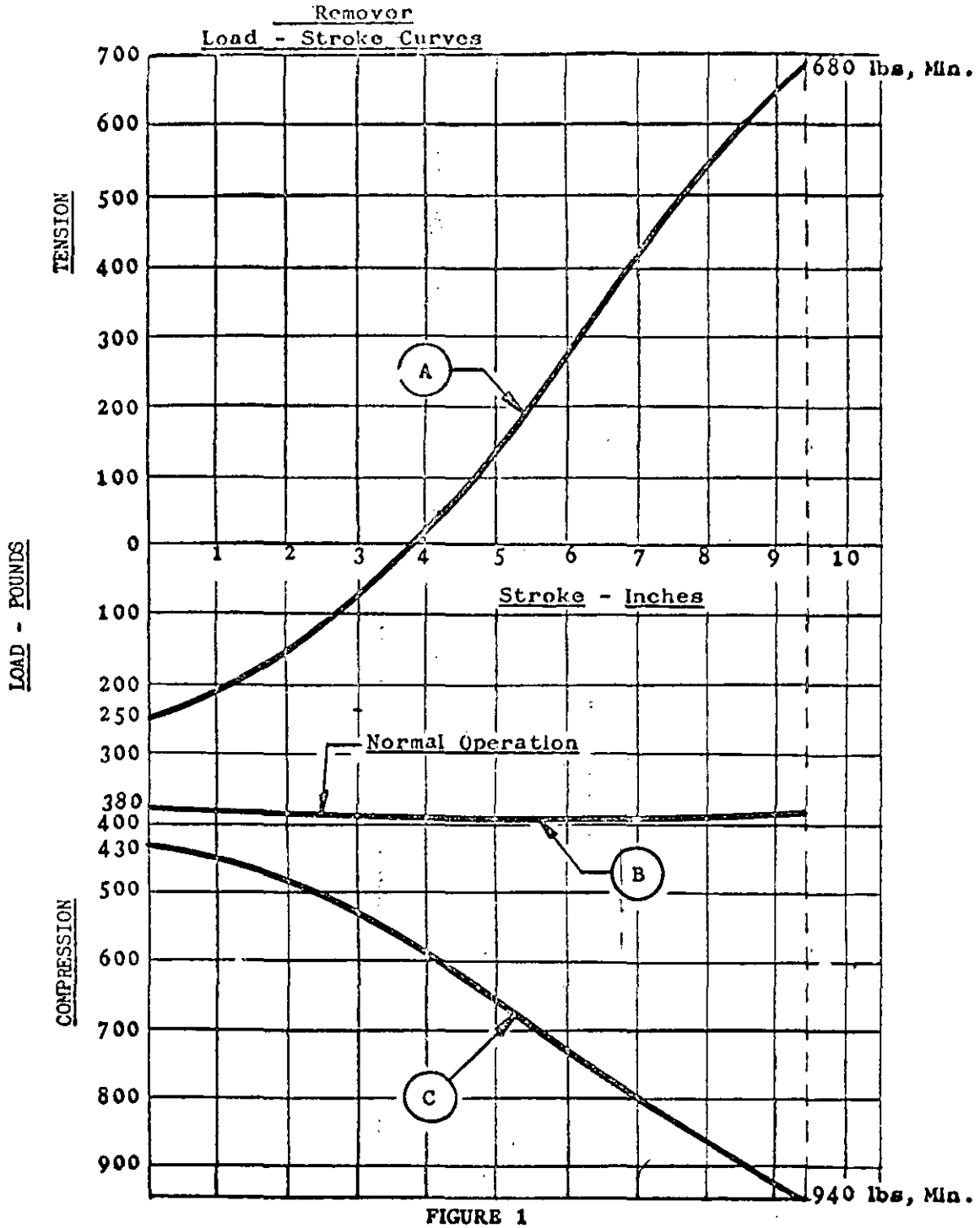


FIGURE 1

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Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-J4000
Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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